WHERE A MORPHISM INTO AN AFFINE SCHEME SENDS A POINT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1001-1017, following the statement at 985-999.

Allowed standard prerequisites (editorial).

- f-maps of sheaves. Sheaves, Definition sheaves-definition-f-map (cross-chapter, cited at ~1011-1012). Supplies the displayed commutative diagram of global-to-stalk maps.

Source reference. Original reference 008J: focused-seeds/geometry/S9-schemes-morphism-into-affine-where-point-goes/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a locally ringed space. Let Y be an affine scheme. Let $f \in \mathrm{Mor}(X, Y)$ be a morphism of locally ringed spaces. Given a point $x \in X$ consider the ring maps*

$$\Gamma(Y, \mathcal{O}_Y) \xrightarrow{f^\sharp} \Gamma(X, \mathcal{O}_X) \to \mathcal{O}_{X,x}$$

Let $\mathfrak{p} \subset \Gamma(Y, \mathcal{O}_Y)$ denote the inverse image of $\mathfrak{m}_x$. Let $y \in Y$ be the corresponding point. Then $f(x) = y$.

Proof. Consider the commutative diagram

$$\begin{array}{ccc} \Gamma(X, \mathcal{O}_X) & \longrightarrow & \mathcal{O}_{X,x} \\ \uparrow & & \uparrow \\ \Gamma(Y, \mathcal{O}_Y) & \longrightarrow & \mathcal{O}_{Y,f(x)} \end{array}$$

(see the discussion of f-maps below Sheaves, Definition 008J). Since the right vertical arrow is local we see that $\mathfrak{m}_{f(x)}$ is the inverse image of $\mathfrak{m}_x$. The result follows. □

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